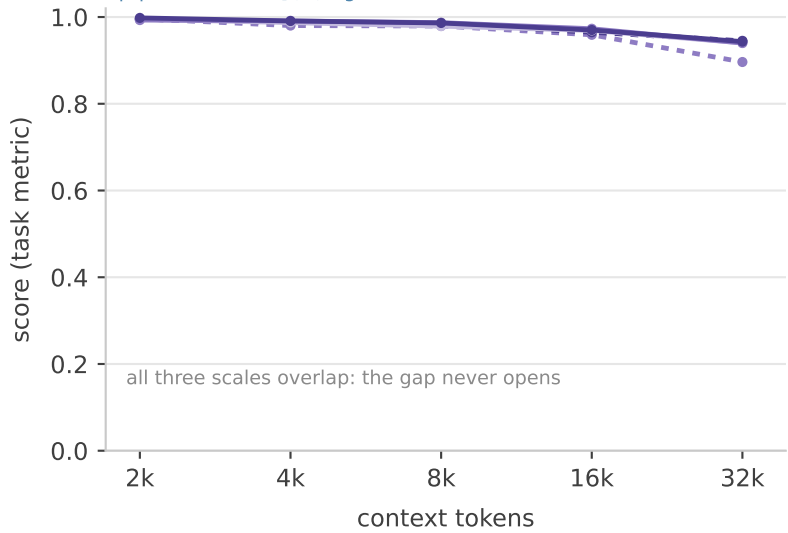


Scaling 0.8B → 4B does not close the chunked-attention gap on high-CTC tasks

Qwen3.5 at three parameter counts, identical training shards and identical evaluation ladders; solid = full attention, dotted = chunked. The gap moves in both directions with scale (contradiction 35%→26%, qdmatch 73%→85%) but stays large either way.

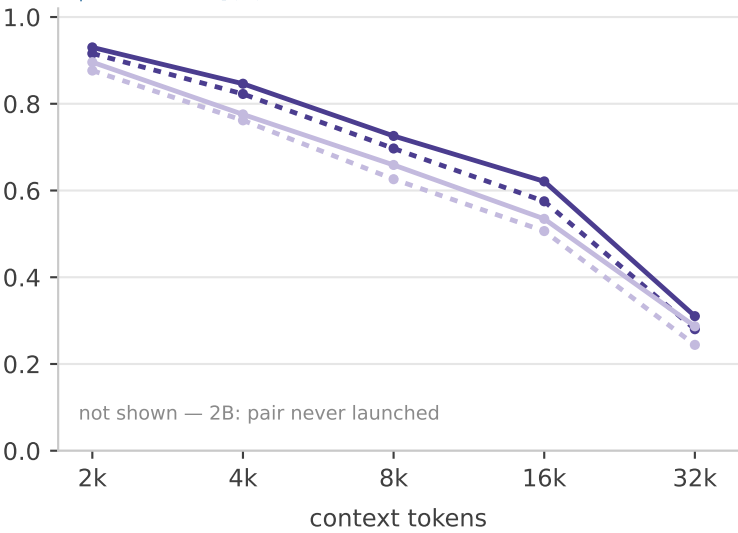
(a) HotpotQA

hpqa · low CTC $O_T(N)$ · gold-ID F1



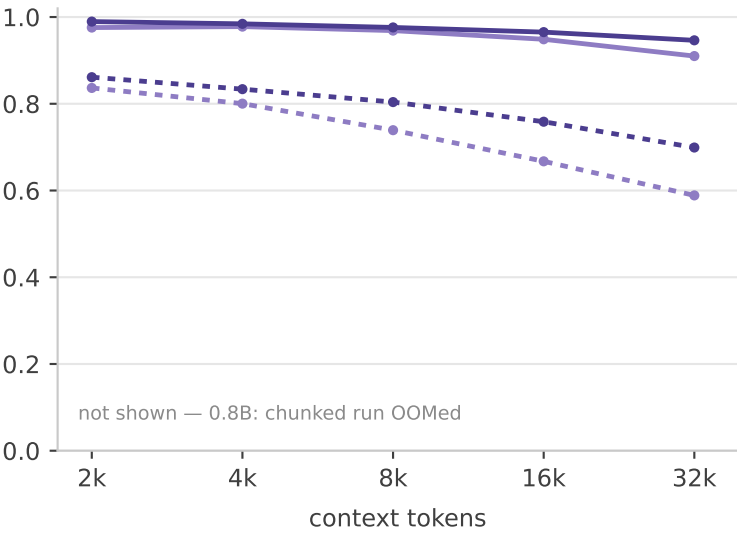
(b) BEIR FiQA

fiqa · low CTC $O_T(N)$ · set-F1



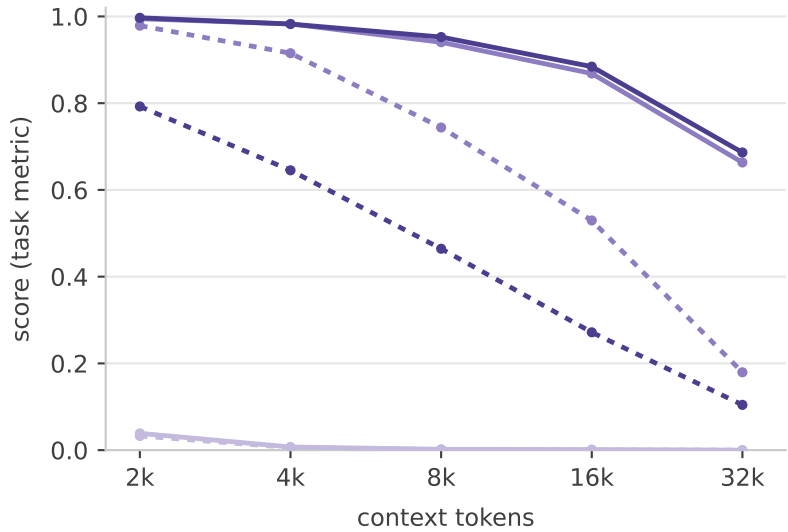
(c) contradiction

contradiction · high CTC $O_T(N^2)$ · set-F1



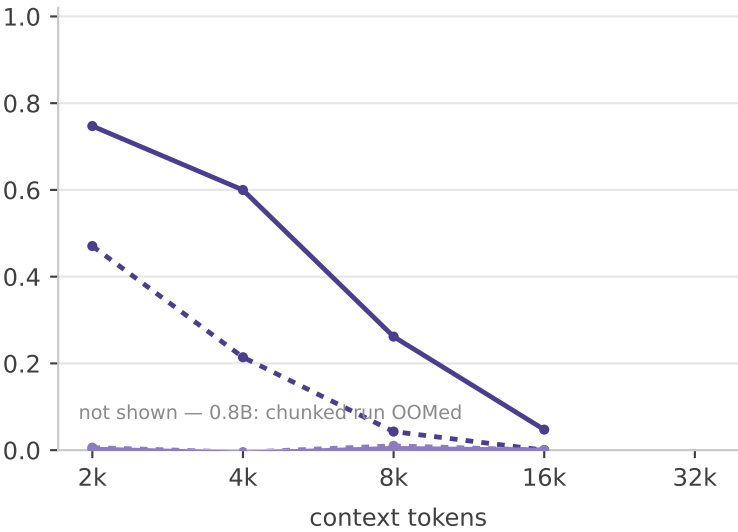
(d) qdmatch (NQ)

qdmatch_nq · high CTC $O_T(N^2)$ · pair-F1



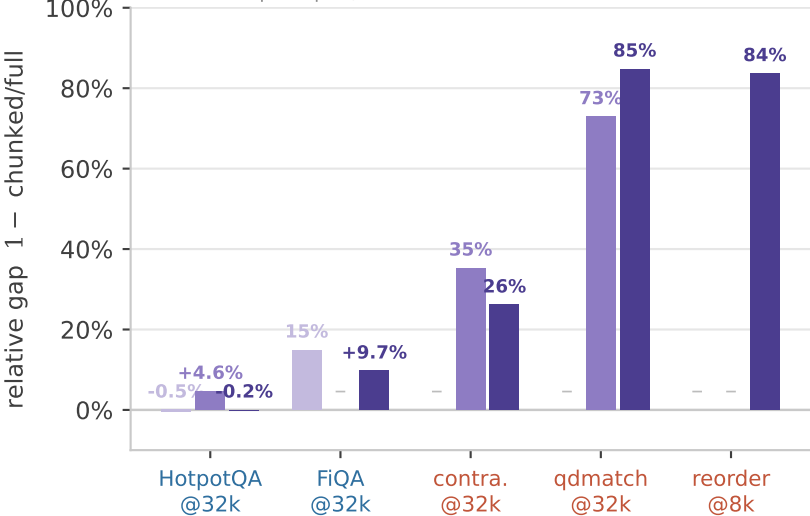
(e) reordering

reorder · high CTC $O_T(N^2)$ · Kendall tau



(f) the gap at the top rung

“—” = no complete pair, or the full arm is under the 0.10 floor where the ratio is noise



Excluded runs (14 of 72 exported rows):

- 0.8B contra_real — 0.8B x chunked-mix x seq_len 40960 OOMs on both saturn and jupiter while 2B/4B and 0.8B@33792 train fine; four hypotheses tested and refuted, cause still unknown --...
- 2B fiqa — fiqa was only ever run at 0.8B and 4B; the 2B pair was not launched
- 0.8B reorder — same unexplained 0.8B x chunked-mix x 40960 OOM as contradiction